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Roll No

CE-8001-CBGS

B.E. VIII Semester

Examination, December 2020

Choice Based Grading System (CBGS)

Advanced Structural Design -II (Steel)

Time : Three Hours

Maximum Marks : 70

Note: i) Attempt any five questions. All questions carry equal marks.

ii) Use of steel table is allowed.

iii) Use of calculator is allowed.

1. A deck type welded plate girder Railway bridge is to constructed for a broad gauge single track on main line. Effective span=20m, c/c distance between the plate girder is 2m , dead load on each girder is (220L+600) N/m, Dead load of track with sleeper=6800 N/m. Design the super structure of the bridge with welded plate girder.
2.
 - a) What are different types of truss bridges? Explain the design principles of optimum depth of truss-girder.
 - b) Explain the various types of bridge bearing and discuss Elastomeric Bearing for bridge structures in detail.
3.
 - a) Distinguish between bunker and silo with the Help of diagram.
 - b) Explain factors affecting the design of bins.
4. Explain the lining material used for chimney.

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5. Design a railway bridge for following data:
Type of bridge - Deck Type Plate Girder Bridge
Span - 16m between centres of bearing
Gauge - Broad, Single track, main line
Distance between centres of plate girders = 2 m
6. Design a rectangular steel overhead tank to hold 1,50,000 litres of water. The height is 2.2m and depth of water is 2m. Steel plates are available in 2m and 1.25m width and of any thickness.
Design:
 - a) The bottom plate
 - b) The bottom tee cover
7. What are the types of steel stacks? Explain the forces acting on it?
8. Explain the following:
 - a) Shape Factor
 - b) Conditions for Plastic Analysis
